Telco Customer Churn Analysis Dashboard

1. Project Overview

This project focuses on analyzing customer churn behavior in a telecom company using Power BI and SQL. The objective is to identify key factors contributing to customer churn and to provide actionable insights that help improve customer retention and business performance.

The dashboard provides a comprehensive view of customer distribution, churn patterns, revenue impact, and customer segmentation.

2. Objectives

- To analyze customer churn rate and identify high-risk segments
- To understand the impact of churn on overall revenue
- To identify key factors such as contract type, tenure, and services affecting churn
- To provide actionable insights for reducing churn and improving retention

3. Tools & Technologies Used

- **SQL** → Data extraction, transformation, and querying
- **Power BI** → Data visualization and dashboard creation
- **Power Query** → Data cleaning and preprocessing
- **DAX (Data Analysis Expressions)** → Creating calculated measures and KPIs

4. Dataset Description

The dataset contains telecom customer information including:

- Customer ID
- Gender
- Tenure
- Contract Type
- Internet Service
- Payment Method
- Monthly Charges
- Total Charges
- Churn (Yes/No)

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5. Data Cleaning & Preparation

- Handled missing values in Total Charges
- Converted data types (e.g., Customer ID as text)
- Created calculated columns such as Tenure Groups
- Structured data into fact and dimension tables for better modeling

6. Key Metrics (KPIs)

- Total Customers
- Total Revenue
- Churn Rate (%)
- Active Monthly Revenue
- Revenue Lost due to Churn

7. Dashboard Structure

Page 1: Executive Overview

- High-level KPIs (Customers, Revenue, Churn Rate)
- Customer distribution by contract and tenure
- Churn distribution using donut chart
- Gauge chart to compare churn vs target
- Customer Distribution by Behaviours

Page 2: Churn Analysis

- Churn Rate by Contract Type
- Churn Rate by Tenure Group
- Churn by Internet Service
- Churn by Payment Method (heatmap)
- Decomposition Tree for root cause analysis

Page 3: Business Impact

- Top 5 high-value customers
- Customer distribution across tenure groups
- Total Revenue by Contract
- Revenue by Payments

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- Revenue lost due to churn
- Waterfall chart showing revenue flow
- Key insights and recommendations

8. Key Insights

- The overall churn rate is **26.54%**, which is significantly higher than the acceptable benchmark (~10%)
- Month-to-month contract customers contribute the most to churn
- New customers (low tenure) are more likely to churn
- Long-term customers generate more stable and higher revenue
- Two-year contracts generate the most revenue ($6.3M) and are most stable.
- Active customers contribute $13M, churned customers account for $3M loss.

9. Business Recommendations

- Introduce incentives for long-term contracts
- Improve onboarding experience for new customers
- Focus on high-risk segments (monthly and low-tenure customers)
- Enhance service quality for high churn categories
- Implement targeted retention campaigns

10. Conclusion

The analysis highlights that customer churn is a major concern for the business, significantly impacting revenue and growth. By focusing on high-risk customer segments and improving retention strategies, the company can reduce churn and increase long-term profitability.

This project demonstrates strong skills in data analysis, visualization, and business problem-solving using Power BI and SQL.

11. Future Scope

- Add time-series analysis if historical data is available
- Build predictive churn models using machine learning
- Integrate real-time dashboards for continuous monitoring